

Energy Shocks and the Green Adoption Channel in a Small Open Economy

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Abstract

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of [Auclert et al. \(2024\)](#). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks under the two fiscal responses studied in the recent literature: a retail price cap ([Langot et al., 2026](#)) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption ([Bayer et al., 2026](#)). The price cap stabilises output but mechanically eliminates the green adoption margin, while the transfer delivers comparable stabilisation and leaves adoption intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. This is our contribution and, we think, its natural boundary: the three closest papers—[Auclert et al. \(2024\)](#), [Bayer et al. \(2026\)](#) and [Langot et al. \(2026\)](#)—all hold the household’s energy technology fixed, and we add the one margin they omit rather than the two-country or labour-market machinery they contain. The transition loss is linear in the cap’s intensity under a price shock and convex under a supply shock. As a secondary observation, the *sign* of the cap’s private-welfare effect flips with the elasticity of energy supply, matching the sign each of [Langot et al. \(2026\)](#) and [Bayer et al. \(2026\)](#) obtains in its own closure; we read this as locating one axis of their contrast, not as nesting their models. The policy *ordering* is robust to calibration; the *magnitude* of the adoption channel, and even the sign of its output effect under a price shock, depend on the calibrated switching cost and taste scale, which we report and discuss.

1 Introduction

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. [Auclert et al. \(2024\)](#) study a small open economy and stress the negative externality when all importers subsidise simultaneously. [Bayer et al. \(2026\)](#) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. [Langot et al. \(2026\)](#) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well. All three treat the household’s energy technology as fixed.

We relax exactly that margin, and only that margin. Households can adopt a green durable that insulates them from the fossil price, at a resource cost. The energy shock raises the return to adoption, so the crisis itself can accelerate the transition—unless policy removes the price signal that drives it. This is the contribution: an endogenous green-adoption margin, absent from all three references. We deliberately do *not* try to nest the two-country spillovers of [Bayer et al. \(2026\)](#) or the labour-cost employment channel of [Langot et al. \(2026\)](#). A small open economy with an exogenous foreign block contains neither, and competing on that machinery is not the point; our single comparative advantage is the margin the literature leaves out, and we play only that.

Because the price-versus-supply distinction turns out to matter for the adoption margin, we do let the energy-supply elasticity vary: with $\varepsilon^E = \infty$ the economy is a price taker (elastic

supply, as in Langot); with ε^E finite the quantity is fixed and the world price clears the market (as in Bayer). Both closures are reported below. This is a modelling convenience for our own question, not a claim to reconcile the two papers: the supply elasticity is one dimension along which they differ, and not the one that carries their headline mechanisms (Section 6).

2 Model

2.1 Conventions and notation

Time is quarterly. A subscript t is suppressed where unambiguous; $x(-1)$ and $x(+1)$ denote lag and (expected) lead. A bar denotes a steady-state value, e.g. $\bar{p}^E$. All *relative prices* carry the subscript “/P” where P is the CPI, e.g. $P_B^E \equiv P_{B,t}^E/P_t$; these are the objects the CES demand system uses.

Unit of account. The unit of account is the domestic good. The household reads, solves in units of account, and returns assets converted to CPI units by $A^{CPI} = A p^{\text{num}}$, with

$$p^{\text{num}} = p_{H/P}, \quad 1 + r^{\text{num}} = (1 + r) \frac{p_{H/P}(-1)}{p_{H/P}}, \quad atw^{\text{num}} = \frac{atw}{p_{H/P}}. \quad (1)$$

Objects contractually denominated in the CPI—fiscal transfers and the switching cost ψ_g —are divided by p^{num} on the way into the budget. Taking the domestic good rather than the CPI as unit of account makes the resource constraint independent of how the CPI weights brown and green energy.

2.2 Households

There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass, with β_i (Section 3); within a type, households face idiosyncratic labour productivity and hold a discrete durable technology.

2.2.1 Idiosyncratic productivity and the durable state

Labour productivity e follows a Rouwenhorst discretisation of an AR(1) in logs, $\log e' = \rho_e \log e + \varepsilon$, with n_e points, persistence ρ_e and innovation s.d. σ_e ; transition matrix Π , mean-one grid.

Each household carries two binary durable indicators: the technology it *operates this period* $g \in \{B, G\}$ and the technology it *held last period* $g_- \in \{B, G\}$. We collapse the pair into a four-point index

$$d \in \{BB, BG, GB, GG\}, \quad (\text{now, before}) = \{(B, B), (B, G), (G, B), (G, G)\}. \quad (2)$$

Define the indicators

$$\text{IS_GREEN}(d) = \mathbb{I}[g = G] = (0, 0, 1, 1), \quad \text{PAYS_SWITCH}(d) = \mathbb{I}[g = G, g_- = B] = (0, 0, 1, 0). \quad (3)$$

The switching cost ψ_g is incurred exactly in the transition $g_- = B \rightarrow g = G$ (state GB). Brown is *absorbing*; a green durable *breaks down* to brown at rate δ_g . The breakdown Markov (applied in the **dep** stage, rows = state this period, columns = state next period) is

$$M_{\text{dep}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ \delta_g & 0 & 0 & 1 - \delta_g \\ \delta_g & 0 & 0 & 1 - \delta_g \end{pmatrix}. \quad (4)$$

On the equilibrium path BG carries zero mass (green is lost only through breakdown, recorded as $g_- = B$), so the effective state is three-point. In steady state a permanent green share is sustained by a flow of switchers,

$$D^{\text{sw}} = \delta_g D^G. \quad (5)$$

Forward stage order within a period is `[dep, prod, durables, consav]`: breakdown, productivity, the discrete adoption choice, and the EGM consumption–savings step. The adoption stage uses the discrete–continuous structure of [Iskhakov et al. \(2017\)](#).

2.2.2 Preferences and the type-specific bundle price

Flow felicity is CRRA over a consumption bundle c with disutility of labour handled at the union level (Section 2.4.1):

$$u(c) = \begin{cases} \log c & \text{eis} = 1 \\ \frac{c^{1-1/\text{eis}}}{1-1/\text{eis}} & \text{eis} \neq 1. \end{cases} \quad (6)$$

The bundle c is a CES nest of energy against a home/foreign composite, with ARS parameters (α_E, η_E) . Because a green adopter buys energy at P_G^E and a brown household at P_B^E , each durable type faces its own energy price and hence its own *exact* cost-of-living index:

$$p_d^E = P_B^E + \text{IS_GREEN}(d) (P_G^E - P_B^E), \quad (7)$$

$$p_d = \left[1 + \alpha_E ((p_d^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}) \right]^{\frac{1}{1-\eta_E}} \quad (\eta_E \neq 1), \quad p_d = (p_d^E / P_B^E)^{\alpha_E} \quad (\eta_E = 1). \quad (8)$$

Equation (8) uses the equilibrium CES identity $\alpha_E (P_B^E)^{1-\eta_E} + (1 - \alpha_E) p_{HF/P}^{1-\eta_E} = 1$ (the block `CESprices`, eq. (32)), so *brown* households have $p_d \equiv 1$ at every date (exact nesting). p_d is CPI-relative; the budget uses $p_d^{\text{num}} \equiv p_d / p^{\text{num}}$.

2.2.3 Cash on hand

Let a be beginning-of-period assets (in units of account, on a grid with lower bound $a \geq \underline{a}$, $\underline{a} = 0$). Cash on hand of a household in durable state d with productivity e is

$$\text{coh}(d, e, a) = (1 + r^{\text{num}}) a + atw^{\text{num}} e + \frac{T^{\text{fisc}}}{p^{\text{num}}} - \frac{\psi_g}{p^{\text{num}}} \text{PAYS_SWITCH}(d), \quad (9)$$

with the (CPI-denominated) transfer

$$T^{\text{fisc}} = \epsilon^T + \iota (p^E - \bar{p}^E) \bar{c}_{E,i}^B(e, a) \quad (\text{untargeted lump sum } \epsilon^T + \text{Slutsky transfer}). \quad (10)$$

Here $\iota = \text{insE}$ scales the Slutsky transfer, $\epsilon^T = \text{epsT}$ is an untargeted lump sum, and $\bar{c}_{E,i}^B(e, a)$ is household i 's steady-state *brown* energy demand, collapsed over the durable axis and broadcast identically across durable states (so a switcher keeps the same transfer; the switching margin is undistorted). Both instruments are zero in the baseline. The switching cost enters cash on hand only in state GB .

2.2.4 Discrete adoption choice

In the `durables` stage the household chooses its current technology g given the technology it holds (post-breakdown), by a logit over continuation values with taste scale $\sigma_\epsilon = \text{taste_shock}$.

Feasible transitions carry the flow payoff

$$f(d \mid d_-) = \begin{cases} 0 & \text{stay/renew brown: } (BB \mid BB), (BB \mid BG), \\ 0 & \text{(masked to } -\infty \text{ if } \text{coh}(GB) \leq 0) \text{ switch to green: } (GB \mid BB), (GB \mid BG), \\ 0 & \text{stay/renew green: } (GG \mid GB), (GG \mid GG), \\ -\infty & \text{otherwise (infeasible).} \end{cases} \quad (11)$$

The switching cost itself is already in $\text{coh}(GB)$ through eq. (9). Given post-choice values $V(d)$, the choice probabilities and the expected (pre-shock) value are the standard logit / log-sum-exp:

$$\Pr(d \mid d_-) = \frac{\exp([f(d \mid d_-) + V(d)]/\sigma_\varepsilon)}{\sum_{d'} \exp([f(d' \mid d_-) + V(d')]/\sigma_\varepsilon)}, \quad \tilde{V}(d_-) = \sigma_\varepsilon \log \sum_{d'} \exp\left(\frac{f(d' \mid d_-) + V(d')}{\sigma_\varepsilon}\right). \quad (12)$$

2.2.5 Consumption-savings (EGM)

Conditional on the durable state, the household solves a standard consumption-savings problem with the type-specific bundle price p_d^{num} . The budget, Euler and envelope are

$$p_d^{\text{num}} c + a' = \text{coh}(d), \quad a' \geq \underline{a}, \quad (13)$$

$$u'(c) = \beta_i p_d^{\text{num}} \mathbb{E} V_a(a', e', d') \quad (\text{Euler}), \quad (14)$$

$$V_a = (1 + r^{\text{num}}) u'(c) / p_d^{\text{num}} \quad (\text{envelope}), \quad (15)$$

$$V = u(c) + \beta_i \mathbb{E} V(a', e', d') \quad (\text{value}), \quad (16)$$

solved by the endogenous grid method, with $u'(c) = c^{-1/\text{eis}}$.

2.2.6 Household outputs (hetoutputs)

Given the policy $c(d, e, a)$, the household's *exact* CES demand split by durable type is

$$c_E(d) = \alpha_E (p_d^E / p_d)^{-\eta_E} c, \quad c_{HF}(d) = (1 - \alpha_E) (p_{HF/P} / p_d)^{-\eta_E} c, \quad (17)$$

$$c_E^B(d) = c_E(d) (1 - \text{IS_GREEN}(d)), \quad c_E^G(d) = c_E(d) \text{IS_GREEN}(d). \quad (18)$$

(Prices in (17) are CPI-relative; the denominator is p_d , not p_d^{num} .) The block also returns $d_G(d) = \text{IS_GREEN}(d)$, $d_{sw}(d) = \text{PAYS_SWITCH}(d)$, the MPC, and flow utility $u(c)$.

2.3 Aggregation

For each type i , SSJ aggregates the hetoutputs over the stationary distribution Λ_i into $C_i, A_i, \text{MPC}_i, D_i^G, D_i^{\text{sw}}, C_E^B$ and the model averages equally across the three types:

$$X = \frac{1}{3} \sum_i X_i, \quad X \in \{C, A, \text{MPC}, D^G, D^{\text{sw}}, C_E^B, C_E^G, C_{HF}, \dots\}, \quad \beta = \frac{1}{3} \sum_i \beta_i. \quad (19)$$

Assets in units of account are converted to CPI units, $A^{\text{CPI}} = A p^{\text{num}}$. Aggregate demands for the home good, the foreign good and energy are then built from the household aggregates (block `hh_outputs_dur`), applying on top only the inner home/foreign nest, common to all households:

$$c_E = C_E^B + C_E^G, \quad c_H = (1 - \alpha_F) p_{H/HF}^{-\eta} C_{HF}, \quad c_F = \alpha_F p_{F/HF}^{-\eta} C_{HF}. \quad (20)$$

2.4 Firms and domestic production

A competitive final domestic good is produced from labour. With productivity Z and a constant gross markup μ_{ss} ,

$$y = Z n, \quad \text{gdp} = Z n, \quad \text{btw}_n = w n, \quad \text{atw}_n = (1 - \tau^Y) w n, \quad (21)$$

where y is home-good output, n hours, w the real wage. Monopolistic profits are capitalised into the mutual fund:

$$\text{div} = (\mu_{ss} - 1) w n, \quad J = \text{div} + \frac{J(+1)}{1 + r^{\text{ante}}}, \quad j = \frac{J(+1)}{1 + r^{\text{ante}}}. \quad (22)$$

2.4.1 Wage setting (sticky wages)

A union sets the nominal wage subject to Calvo frictions θ_w , giving a wage Phillips curve. With $\kappa_w = (1 - \theta_w)(1 - \beta\theta_w)/\theta_w$, labour-disutility shifter φ (`vphi`), Frisch elasticity `frisch`, and a real-wage-rigidity term controlled by w_{BG} ,

$$\pi^w(1 + \pi^w) = \beta \pi^w(+1)(1 + \pi^w(+1)) + \frac{\kappa_w}{\varphi n^{1/\text{frisch}}} \Xi, \quad (23)$$

$$\Xi = \varphi n^{1/\text{frisch}} - \frac{\text{atw } C^{-1/\text{eis}}}{\mu_{ss}} - w_{BG} \underbrace{\frac{(w - \bar{w}) w \bar{C}^{-1/\text{eis}} \bar{n}}{\mu_{ss} n \bar{w}}}_{\text{real-wage rigidity}}, \quad (24)$$

and the nominal and real wage evolve as $W = (1 + \pi^w)W(-1)$, $w = W/P$. Here $\text{atw} = \text{atw}_n/n$ is the after-tax wage per hour.

2.5 Prices

All retail prices are relative to the CPI P ; Q is the real exchange rate (price of the foreign basket in home units), P_F^* , P^{E*} are exogenous world prices.

Domestic good.

$$P_H = \frac{\mu_{ss}}{Z} W, \quad p_{H/P} = P_H/P, \quad \pi^H = P_H/P_H(-1) - 1, \quad w = W/P. \quad (25)$$

Imported foreign good (Calvo θ_F). With $\kappa_F = (1 - \theta_F)(1 - \beta_F\theta_F)/\theta_F$, $\beta_F = 1/(1 + \bar{r}^{\text{ante}})$,

$$\kappa_F \left(\frac{QP_F^*}{p_{F/P}} - 1 \right) + \beta_F \pi^F(+1)(1 + \pi^F(+1)) - \pi^F(1 + \pi^F) = 0, \quad p_{F/P} = P_F/P. \quad (26)$$

Energy retail price and the price cap (Calvo θ_E). The wholesale/retail brown energy price $p^E = P_E/P$ satisfies a NKPC, and the *household* brown price applies the cap:

$$\kappa_E \left(\frac{QP^{E*}}{p^E} - 1 \right) + \beta_E \pi^E(+1)(1 + \pi^E(+1)) - \pi^E(1 + \pi^E) = 0, \quad (27)$$

$$\text{with } \kappa_E = (1 - \theta_E)(1 - \beta_E\theta_E)/\theta_E \quad (28)$$

And,

$$P_B^E = (1 - \tau^E) p^E + \tau^E \bar{p}^E, \quad (29)$$

$$\text{so when } \tau^E = 1 \Rightarrow P_B^E \text{ pinned at } \bar{p}^E. \quad (30)$$

Green energy price. With green/brown ratio $\varrho = \text{pE_g_ratio}$, the green price is fixed relative to the pre-crisis brown price:

$$P_G^E = \varrho \bar{p}^E. \quad (31)$$

CPI (brown-anchored; “Option C”). The energy leg of the CPI is anchored on the brown price alone. With $\alpha = \alpha_E + (1 - \alpha_E)\alpha_F$, the inner (energy vs home/foreign) and outer (home vs foreign) CES identities that define $p_{HF/P}$ and $p_{H/HF}$ are

$$(1 - \alpha_E) p_{HF/P}^{1-\eta_E} + \alpha_E (P_B^E)^{1-\eta_E} = 1, \quad (32)$$

$$(1 - \alpha_F) p_{H/HF}^{1-\eta} + \alpha_F p_{F/HF}^{1-\eta} = 1, \quad (33)$$

with $p_{F/HF} = p_{F/P}/p_{HF/P}$, $p_{H/P} = p_{H/HF} p_{HF/P}$, $p_H^* = p_{H/P}/Q$. The gap between this measured CPI and the share-weighted (“true”) deflator is computed *ex post*, outside the model:

$$\phi^{1-\eta_E} = 1 + \alpha_E D^G [(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}]. \quad (34)$$

Under the domestic-good numeraire this choice affects only the measured deflator and the monetary rule, not the resource constraint.

Aggregate price level and inflation.

$$P = (1 + \pi)P(-1), \quad \pi^{\text{ann}} = (1 + \pi)^4 - 1, \text{ etc.} \quad (35)$$

2.6 Open economy

Foreign demand for the home good. With foreign demand shifter α^* , trade elasticity γ , and world demand C^* ,

$$c_H^* = \alpha^* (p_H^*)^{-\gamma} C^*. \quad (36)$$

Uncovered interest parity. The real exchange rate solves

$$\frac{Q}{Q(+1)} (1 + r^{\text{ante}}) = 1 + r^*. \quad (37)$$

Importer margins (foreign-owned). Retail importers of the foreign good and of energy are *owned by foreigners* (ARS, §4.1), so their margins over the world price accrue abroad and are *not* capitalised into domestic wealth. They enter the model only by valuing the import bill at the retail (not world) price in the balance of payments below:

$$D_F = (p_{F/P} - QP_F^*)c_F, \quad D_E = (p^E - QP^{E*})c_E. \quad (38)$$

Both margins are zero in steady state ($p_{F/P} = QP_F^*$, $p^E = QP^{E*}$), so this convention leaves the steady state unchanged.

World energy supply (IEA block). World supply is a fixed quantity E^{shock} plus intertemporal stock arbitrage with cost Γ_{arb} :

$$E_{\text{stock}} = \frac{P^{E*}(+1)/(1 + r^*) - P^{E*}}{\Gamma_{\text{arb}}}, \quad E^{\text{supply}} = E^{\text{shock}} + (E_{\text{stock}}(-1) - E_{\text{stock}}). \quad (39)$$

In the baseline the home economy owns *none* of the world energy endowment ($\zeta^E = 0$: a pure importer), so the associated rents are not capitalised into domestic wealth. The endowment-ownership channel ($\zeta^E > 0$, ARS §3.4/§4.1) is retained in the code and used only as a robustness/energy-independence experiment.

Fund return (revaluation). The realised (ex post) return on the fund's claims defines r :

$$1 + r = \frac{J}{j(-1)}. \quad (40)$$

Importers are foreign-owned and the energy endowment is not held domestically ($\zeta^E = 0$), so the only domestically held risky claim is domestic firms: $A^{\text{dom}} = j$, with return $r^{\text{dom}} = r$; A^{dom} enters only the current-account revaluation term below.

Balance of payments. The green margin enters the import booking through two terms,

$$\text{imports}_{\text{dur}} = \psi_g D^{\text{sw}} \quad (\text{switching cost booked as import}), \quad (41)$$

$$\text{energy_gap} = (P_B^E - P_G^E) C_E^G \quad (\text{adopters' import saving}). \quad (42)$$

Exports, imports and net exports are

$$\text{exports} = Q p_H^* c_H^*, \quad (43)$$

$$\text{imports} = p_{F/P} c_F + p^E c_E + \text{imports}_{\text{dur}} - \text{energy_gap}, \quad (44)$$

$$nx = Q p_H^* c_H^* - p_{F/P} c_F - p^E c_E - \text{imports}_{\text{dur}} + \text{energy_gap}. \quad (45)$$

Net foreign assets accumulate with a revaluation term:

$$\text{reval} = (r - r^{\text{ante}}(-1)) A^{\text{CPI}}(-1) - (r^{\text{dom}} - r^{\text{ante}}(-1)) A^{\text{dom}}(-1), \quad (46)$$

$$\text{nfa} = nx + \text{reval} + (1 + r^{\text{ante}}(-1)) \text{nfa}(-1), \quad nx/\text{gdp} = nx/y. \quad (47)$$

2.7 Policy

2.7.1 Monetary policy

The nominal rate rule and the Fisher ex-ante real rate are

$$1 + i = (1 + \bar{r}^*) (1 + \phi_\pi \pi) (1 + \phi_{\pi e} \pi(+1)) + \text{ishock}, \quad r^{\text{ante}} = \frac{1 + i}{1 + \pi(+1)} - 1. \quad (48)$$

The baseline is the ARS *constant real rate* rule $(\phi_\pi, \phi_{\pi e}) = (0, 1) \Rightarrow r^{\text{ante}} = \bar{r}^*$; a standard Taylor rule sets $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$. **ishock** is a monetary policy shock (zero unless a monetary experiment is run).

2.7.2 Fiscal policy

The government runs the two energy-crisis instruments and an untargeted lump sum, all debt financed, with debt stabilised through the distortionary labour tax:

$$\text{Subsidy} = \tau^E (p^E - \bar{p}^E) c_E \quad \text{price cap (cost scales with actual use)}, \quad (49)$$

$$\text{Ttargeted} = \iota (p^E - \bar{p}^E) \bar{C}_E^B \quad \text{Slusky transfer (cost scales with pre-crisis brown use)}, \quad (50)$$

$$\text{Tuntargeted} = \epsilon^T, \quad (51)$$

$$\text{spending} = \text{Tuntargeted} + \text{Ttargeted} + \text{Subsidy}, \quad (52)$$

$$\text{taxation} = \tau^Y w n. \quad (53)$$

The government budget constraint and the debt-stabilising tax rule are

$$B = (1 + r^{\text{ante}}(-1)) B(-1) + \text{spending} - \text{taxation}, \quad (54)$$

$$\tau^Y = \psi_B (B(-1) - \bar{B}), \quad \psi_B = \text{psiB}. \quad (55)$$

In the baseline $\bar{B} = 0$ (no steady-state government debt): crisis outlays are deficit-financed and repaid by raising the labour tax in proportion to the debt's deviation from zero. Government debt B is a genuine transition unknown (it moves in the dynamics) and enters household wealth through asset-market clearing, eq. (57).

2.8 Market clearing and equilibrium

The equilibrium residuals are

$$\text{goods: } c_H + c_H^* - y = 0, \quad (56)$$

$$\text{assets (Walras): } A^{CPI} - \text{nfa} - j - B = 0, \quad (57)$$

$$\text{energy: } \begin{cases} P^{E*} - P^{E*,\text{shock}} = 0 & \varepsilon^E = \infty \text{ (price taker; price shock),} \\ c_E - E^{\text{supply}} = 0 & \varepsilon^E < \infty \text{ (fixed quantity; supply shock).} \end{cases} \quad (58)$$

Equation (57) is a Walras'-law residual (not itself a solved target); it is monitored to machine precision as a consistency check.

Transition unknowns/targets. The linear (SSJ) transition solves for $\{y, \pi, r, \tau^Y, P^{E*}, w\}$ against $\{\text{goods}, \pi \text{ residual}, r \text{ residual}, \tau^Y \text{ residual}, \text{energy}, w \text{ residual}\}$.

2.9 Steady state and shocks

2.9.1 Openness normalisations

With $\zeta^E = 0$ (pure importer) the ARS home-endowment rescaling is inactive, so the normalisations reduce to

$$\alpha_F = \frac{\alpha - \alpha_E}{1 - \alpha_E}, \quad \gamma = \frac{\chi_{\text{tgt}} - (1 - \alpha)(1 - \alpha_F)\eta_E}{(1 - \alpha)\alpha_F + \alpha}, \quad \eta = \gamma, \quad \chi_{\text{tgt}} = 0.3, \quad (59)$$

$$y = 1, \quad Z = \mu_{\text{ss}} = 1.03, \quad \alpha^* = \alpha, \quad E^{\text{shock}} = \alpha_E. \quad (60)$$

2.9.2 Steady-state unknowns/targets

The steady state solves for $\{\varphi, \beta_{\text{max}}, y, \psi_g\}$ against $\{\pi^w \text{ res}, \text{nfa res}, \text{goods}, D^G - 0.05\}$: ψ_g is calibrated to the level target $D_{\text{ss}}^G = 0.05$ and β_{max} closes net foreign assets. In the “no-adoption” counterfactual (`green_block` large) $D^G \equiv 0$ for every ψ_g , so ψ_g is not identifiable and is carried over, fixed, from the matching adoption steady state.

2.9.3 Shocks

$$\text{Brown-price shock (requires } \varepsilon^E = \infty): P_t^{E*,\text{shock}} = \text{size} \cdot \rho^t, \quad \rho = 2^{-1/\text{half-life}}, \quad (61)$$

$$\text{Brown-supply shock (requires } \varepsilon^E < \infty): E_t^{\text{shock}} = \bar{E}^{\text{shock}} - \text{drop} \cdot \bar{E}^{\text{shock}} \text{ for } t < \text{quarters, else } \bar{E}^{\text{shock}}. \quad (62)$$

The baseline price shock is $\text{size} = 1$ (world price doubles on impact), half-life 16; the baseline supply shock is -10% for 6 quarters.

3 Calibration

Parameter	Value	Source / target
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
r, μ_{ss}	0.01, 1.03	Auclert et al. (2024)
β heterogeneity (3 types)	spread 0.06	fixed; implied MPC ≈ 0.30
δ_g green breakdown rate	0.05	assumption (5 yr life)
$\varrho = P_G^E/P_B^E$ steady-state gap	0.80	heat-pump / EV delivered-cost ratio
ψ_g switching cost	0.538	targets $D_{ss}^G = 0.05$
σ_ε taste scale	0.05	calibrated jointly with ψ_g
ρ_g pass-through to green price	0	upper bound on the channel

Table 1: Calibration. The steady state is identical across all experiments. ψ_g and σ_ε are calibrated jointly to the level target $D_{ss}^G = 0.05$; σ_ε is not separately pinned down by an adoption-elasticity moment (Section 7).

4 Experiments and results

The oil shocks The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life, as in [Auclert et al. \(2024\)](#). The *supply* shock is a 10% cut in available energy for 6 quarters, after which the world price clears the market endogenously. Impulse responses are reported as 100 $\times$ the level deviation from steady state. Variables whose steady state is one (prices, w , n , and the population shares D^G , D^{sw}) read directly as percent or percentage-point deviations, and C (steady state ≈ 0.997) is within rounding of a percent; y and gdp have steady state ≈ 0.985 , so their entries are level deviations $\times 100$, within about 1.5% of the exact percentage. Energy quantities are a share of the energy steady state (≈ 0.040).

4.1 Constructing the no-adoption counterfactual

Several results below isolate the adoption channel’s contribution as the difference between the full model and a "no-adoption" counterfactual, $\Delta_H \equiv \sum_{t < H} (y_t^{\text{adopt}} - y_t^{\text{no-adopt}})$. The counterfactual is constructed to share the full model’s steady state *exactly*: households still hold the discrete durable-choice margin, and the steady-state green share, switching flow, and switching cost ($\bar{D}^G = 5\%$, D_{ss}^{sw} , ψ_g) are bit-identical to the adoption economy. What differs is the *transition* only: the discrete-choice probabilities are held at their steady-state values throughout the impulse response (the choice stage’s law of motion is the steady-state one, undifferentiated by the shock), so the durable composition does not respond to the crisis, while every other block — prices, wages, the household’s continuous consumption/saving choice, monetary and fiscal policy — responds normally. We refer to this as **common-steady-state**, or *frozen-choice*.

This matters because the alternative construction — shutting the discrete choice altogether by making green durables infeasible, so the counterfactual’s own steady state has $D_{ss}^G = 0$ — compares two economies that differ both in the crisis response *and* in steady-state composition and asset holdings.

Shock	Policy	$y(0)\%$	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption $\rightarrow \sum y$
price	none	-2.75	-61.5	+7.32	8.87	0.00	-8.79
	subsidy	-1.66	-53.0	-3.37	-0.04	53.86	+0.69
	transfer	-0.63	-22.3	+9.09	9.15	49.27	-8.93
supply	none	-1.03	-13.9	+8.86	1.91	0.00	-0.35
	subsidy	-1.72	-45.6	-1.41	-0.03	-18.45	+0.09
	transfer	+0.56	-8.5	+12.00	2.59	17.74	+0.22

Table 2: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin: the difference between the full model and a common-steady-state counterfactual in which the adoption choice does not respond to the shock (Section 2).

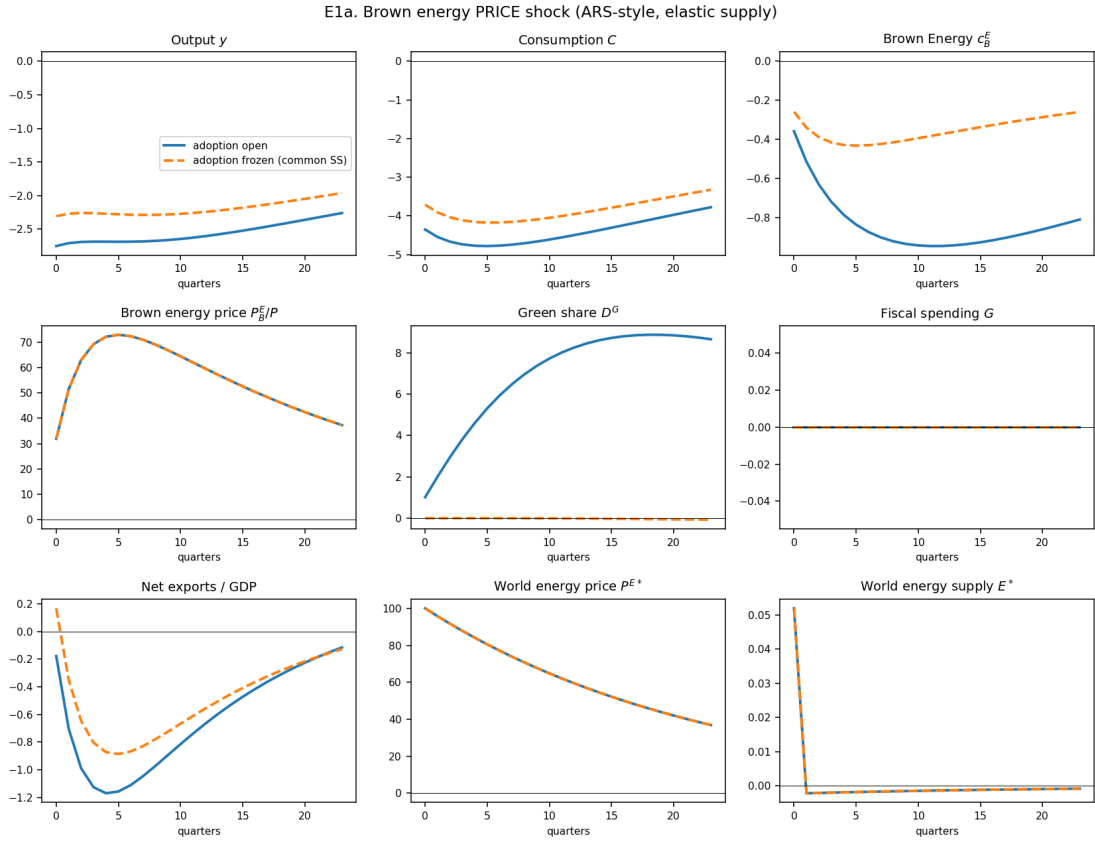


Figure 1: Brown energy *price* shock (elastic supply), adoption open (solid) vs. shut (dashed), no policy.

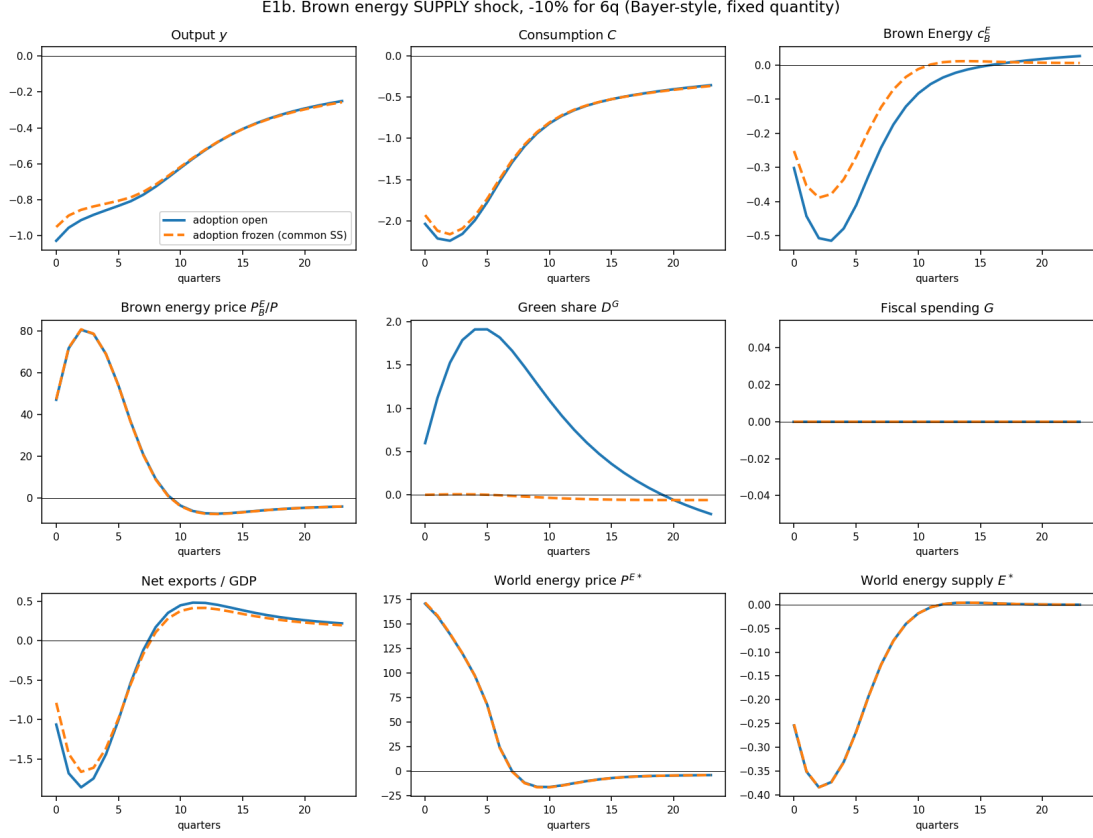


Figure 2: Brown energy *supply* shock (-10% for 6 quarters, fixed quantity), adoption open (solid) vs. shut (dashed), no policy.

4.2 The adoption channel is contractionary

The adoption channel's contribution to cumulative output, $\Delta_H \equiv \sum_{t < H} (y_t^{\text{adopt}} - y_t^{\text{no-adopt}})$, follows a race between two resource flows in the balance of payments (Section 2.6). Adoption imports a **switching cost** $\psi_g D_t^{sw}$ —a resource outflow, *front-loaded* (paid once by each switcher) and scaling with the adoption *volume* D^{sw} —and, against it, a **flow energy saving** $(P_B^E - P_G^E) C_{E,t}^G$ booked as reduced imports, a resource inflow accruing *gradually* over the life of the green durable. Under the common-steady-state counterfactual (Section 4.1) and the import booking, the front-loaded cost dominates the windowed saving at every persistence. Under the fixed-quantity (supply) closure a third, expansionary term appears—adoption lowers fossil demand against a fixed supply, so the clearing price falls and relaxes scarcity for *all* households—which shifts Δ_H upward relative to the price closure. The marginal switcher's steady-state indifference, $\psi_g \approx (\bar{P}_B^E - \bar{P}_G^E) \bar{C}_E^G / (r + \delta_g)$, ties the threshold to the cost ratio ϱ .

4.3 The price cap eliminates the adoption margin

Under the cap, the peak green share response falls to essentially zero under both shocks: the adoption channel is not attenuated, it is switched off. The mechanism is direct: the cap holds P_B^E at its pre-crisis level, and P_B^E is precisely what determines the return to adoption.

E2. Fiscal response to the PRICE shock (adoption margin open)

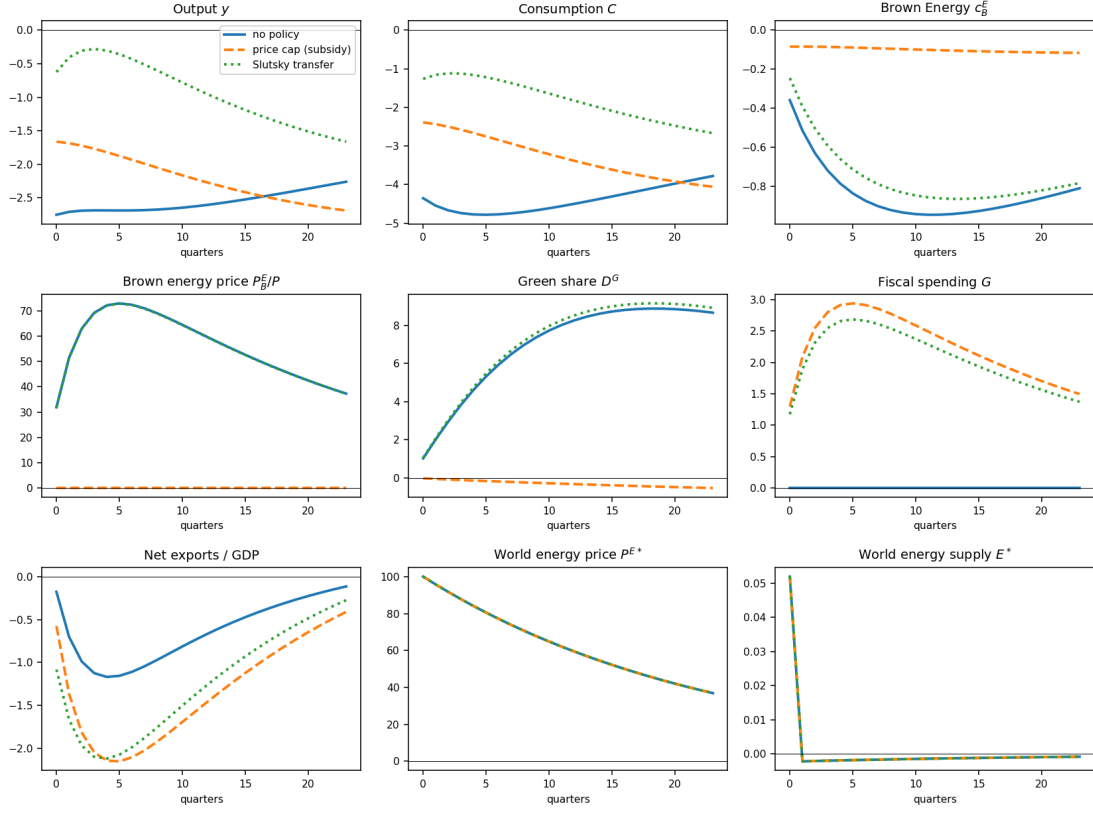


Figure 3: Fiscal response to the *price* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

E2. Fiscal response to the SUPPLY shock (adoption margin open)

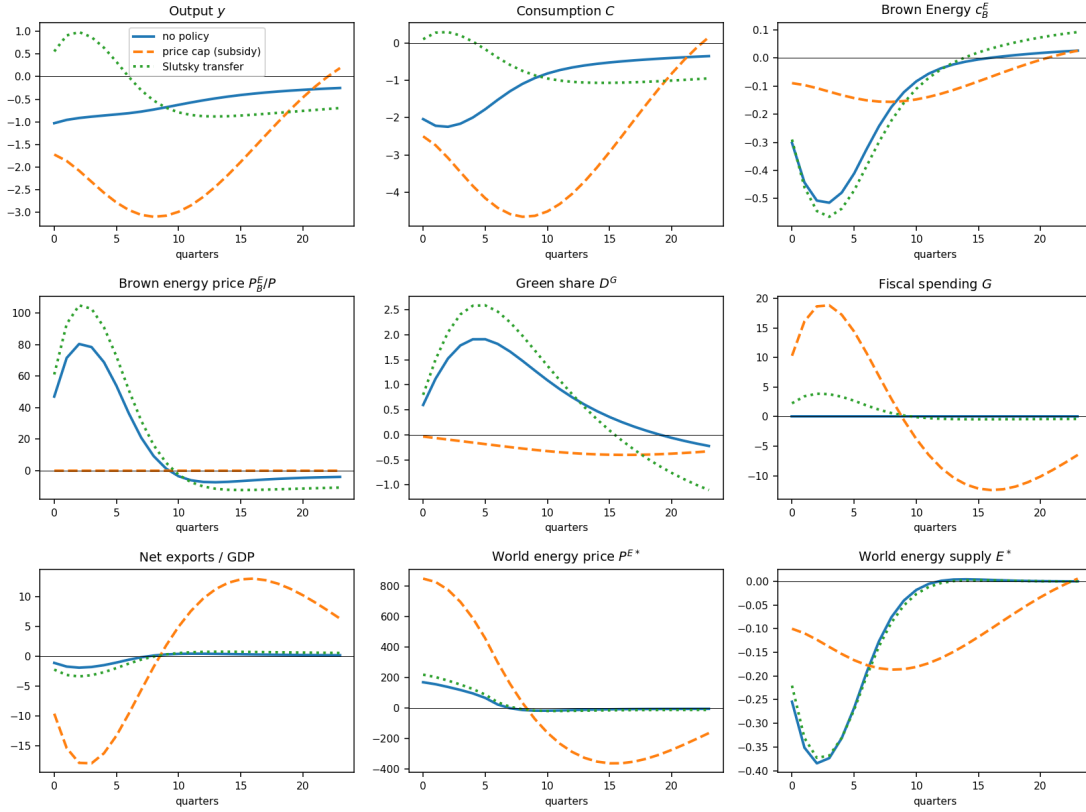


Figure 4: Fiscal response to the *supply* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

4.4 The transfer preserves adoption, at a cost to output

The Slutsky transfer leaves the green share response essentially unchanged while delivering more output stabilisation than the cap, at similar fiscal cost: subsidies destroy the transition margin, transfers do not. This preservation is a direct consequence of the transfer’s lump-sum indexing (Section 2.6): because it does not condition on the household’s current durable, switching forfeits nothing and the relative-price incentive to adopt is left intact.

4.5 Targeting energy transfers is regressive

The Slutsky transfer of Section 4.4 indexes the lump sum to each household’s pre-crisis brown-energy use. Replacing it with an *untargeted* transfer of the *same aggregate envelope*—the identical outlay (49.3 vs. 49.3 over $H = 24$), paid flat across the distribution rather than indexed—almost halves the welfare loss, from a CEV of -0.82% to -0.32% (Table 5). The gain is entirely distributional. By type, the flat transfer moves the impatient (high-MPC) household from -0.68% to $+0.10\%$ —it is made *better off* than in the no-shock steady state—while the patient household bears more of the burden (-0.77%); the Slutsky transfer, by contrast, spreads the loss almost evenly (-0.68 , -0.89 , -0.88).

The mechanism is that energy-indexed targeting is regressive *in this model*. With homothetic CES energy demand the energy share is constant across the wealth distribution, so brown-energy use rises with total consumption, hence with permanent income and patience. Indexing the transfer to pre-crisis energy use therefore hands the largest cheques to the low-MPC, wealthy households, exactly where a euro buys the least insurance. A flat transfer reaches the high-MPC households the crisis hits hardest. This speaks directly to the design of the 2022–23 European

energy-price “brakes,” which were indexed to historical consumption: within the model, the indexing is the source of the inefficiency, not a refinement of it.

We flag the caveat plainly. The result rests on the homothetic energy nest: if energy were a necessity with a declining Engel curve—the empirically standard case—low-income households would spend a *larger* share on energy and the sign could reverse. The clean statement is conditional: *under constant energy shares, energy-indexed targeting is dominated by an equal-value flat transfer*.

4.6 Fixed supply makes the cap destructive

Under the supply shock the cap yields cumulative output of -46.9 against -8.2 with no policy: with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief. This survives the addition of an adoption margin. It is also precisely the configuration in which [Langot et al. \(2026\)](#), assuming elastic supply, find the cap effective: our elastic-supply column agrees with them (-20.9 vs. -30.1 for cap vs. no policy under the price shock).

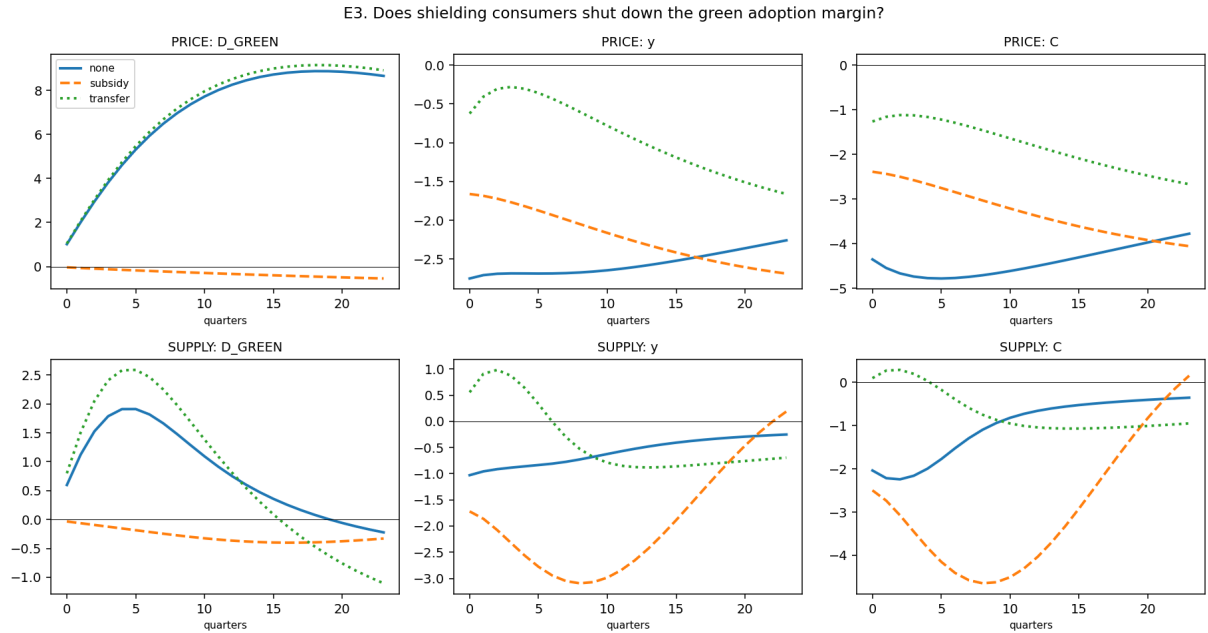


Figure 5: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

4.7 The cap is a dial, not a switch

Table 2 used $\tau^E = 1$, a complete price freeze. Table 3 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in τ^E . Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic—but, as the next result shows, much more damaging to everything else.

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	$\sum c_E$	fiscal	CEV %
price	cap $\tau^E = 0.00$	8.871	-61.46	7.32	-8.94	0.00	-1.8710
	cap $\tau^E = 0.25$	6.547	-59.36	4.68	-7.50	13.40	-1.6917
	cap $\tau^E = 0.50$	4.218	-57.25	2.01	-6.06	26.85	-1.5112
	cap $\tau^E = 0.75$	1.894	-55.15	-0.67	-4.62	40.34	-1.3296
	cap $\tau^E = 1.00$	-0.036	-53.04	-3.37	-3.17	53.86	-1.1469
	transfer	9.146	-22.27	9.09	-6.43	49.27	-0.8162
supply	cap $\tau^E = 0.00$	1.911	-13.94	8.86	-2.40	0.00	-0.6280
	cap $\tau^E = 0.25$	1.754	-16.89	7.99	-2.40	4.61	-0.6694
	cap $\tau^E = 0.50$	1.503	-21.48	6.71	-2.41	11.01	-0.7355
	cap $\tau^E = 0.75$	1.031	-29.64	4.52	-2.43	18.36	-0.8545
	cap $\tau^E = 1.00$	-0.033	-45.55	-1.41	-2.79	-18.45	-1.0843
	transfer	2.588	-8.49	12.00	-2.40	17.74	-0.2881

Table 3: Dose-response in the price-cap intensity τ^E , against the Slutsky transfer benchmark (full compensation, $\iota = 1$).

4.8 The welfare effect of the cap flips sign with the supply elasticity

Under the price shock, tightening the cap *raises* welfare monotonically, from $-X\%$ to $-X\%$; under the supply shock it *lowers* welfare monotonically, from $-X\%$ to $-X\%$. The sign of the cap’s welfare effect therefore hinges on the supply elasticity. [Langot et al. \(2026\)](#), assuming elastic supply, find the French shield effective; [Bayer et al. \(2026\)](#), assuming fixed union-wide supply, find subsidies destructive. Our two closures reproduce the *sign* each obtains, which locates one source of the contrast in a single parameter. We do not claim to reconcile the two papers fully: Langot’s result also runs through a labour-cost channel and Bayer’s through cross-border spillovers in a monetary union, and our small open economy nests neither. The narrower point we establish is that, once the supply elasticity is fixed, the direction of the cap’s welfare effect is determined—so part of the empirical disagreement reduces to measuring that elasticity at the relevant horizon.

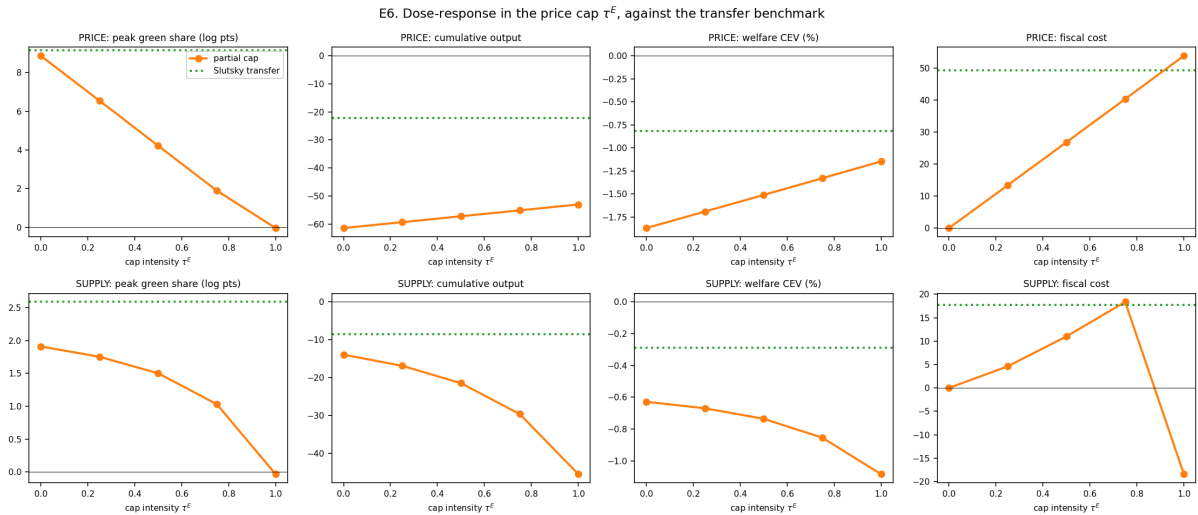


Figure 6: Dose-response in τ^E (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

Private welfare and the transition are distinct. The CEV above is a *private* household-welfare measure: it values consumption and leisure along the transition and prices no climate or

transition externality. Two results should therefore be kept separate. First, on private welfare alone the Slutsky transfer dominates the cap under *both* shocks ; the case for the transfer over the cap does not rest on the transition. Second, and separately, the transfer preserves the green adoption margin that the cap eliminates (Section 4.3)—a positive statement about the energy transition, not a welfare ranking. We deliberately do not aggregate the two into a single social-welfare number: that would require a social value of adoption (a carbon price times avoided brown-energy use) that we do not model. A positive such value only reinforces the private ranking; it cannot reverse it here, since the transfer already dominates.

4.9 Distributional incidence

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	−1.87	−2.37	−2.05	−1.20
cap $\tau^E = 0.5$	−1.51	−1.71	−1.65	−1.17
cap $\tau^E = 1$	−1.15	−1.05	−1.25	−1.14
transfer	−0.82	−0.68	−0.89	−0.88

Table 4: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs. Private household welfare only.

Without policy the impatient lose far more than the patient.

4.10 Monetary policy: Taylor rule versus a constant real rate

The experiments so far hold monetary policy at the ARS constant-real-rate baseline. A 100-basis-point (annualised) monetary tightening isolates the role of the rule itself. Figure 7 contrasts the constant-real-rate baseline with a standard Taylor rule $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$. The qualitative response is the textbook one—output and consumption fall, inflation falls—and it is *not* materially reshaped by the adoption margin: at a 5% green steady state the durable-choice channel is too small to move the aggregate monetary transmission, which is why we keep the constant-real-rate rule as the baseline for the energy-shock experiments and report the Taylor rule only as this robustness contrast. The adoption margin matters for how the economy absorbs an *energy* shock, not for how it transmits a monetary one.

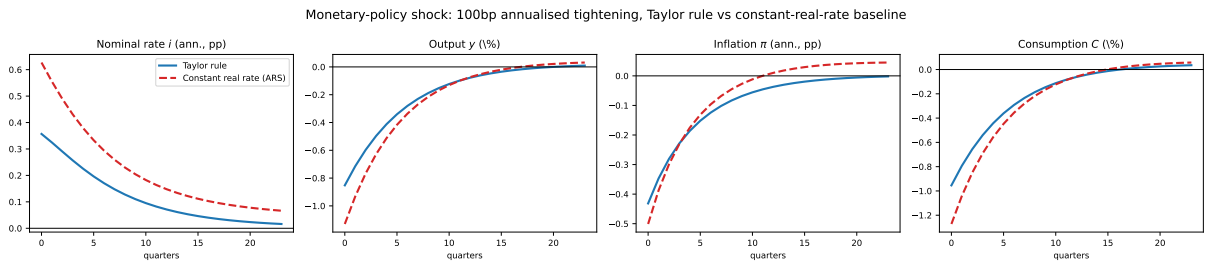


Figure 7: Monetary-policy shock (100bp annualised tightening): Taylor rule (solid) versus the constant-real-rate baseline (dashed). Nominal rate, output, inflation, consumption.

5 Ex ante greening versus ex post insurance

The instruments in Section 4.3–4.4 are all *ex post*: they are deployed once the price shock has landed. Because the adoption margin responds to the same relative price the crisis moves, a natural alternative is to act *ex ante*—to lean on the margin in the steady state with a permanent

carbon tax τ_b on brown energy, leaving the economy greener and, one might hope, less exposed when the shock arrives. This section prices that hope. The answer is that ex ante greening is a poor substitute for ex post insurance, and the reason is intrinsic to the channel this paper adds: the adoption margin is an endogenous shock absorber, and pre-greening spends it.

Table 5 collects the instruments on one axis. The welfare metric is the total consumption-equivalent variation (CEV) of each full scenario relative to the common no-tax steady state, by discount-factor type and averaged, computed ex post from the cached impulse response (`welfare.cev_total`); a permanent policy enters through the standing gap between its steady state and the baseline, a crisis-time policy through the transition alone, so the two are directly comparable.

Scenario	$y(0)\%$	$\sum C\%$	peak D_G	gross fisc.	CEV%
Laissez-faire	-2.75	-105.6	8.87	0.0	-1.871
Ex-post cap	-1.66	-78.9	-0.04	53.9	-1.147
Slutsky transfer	-0.63	-43.4	9.15	49.3	-0.816
Flat transfer	-0.31	-37.3	9.10	49.3	-0.317
Green subsidy	-3.19	-118.5	18.02	2.1	-2.032
Ex-ante ETS ($\tau_b = 0.1$)	-2.85	-106.5	14.98	0.1	-1.852

Table 5: Price-shock policy responses (import booking). $y(0)$ impact output, $\sum C$ cumulative consumption over $H = 24$, peak green share, gross fiscal disbursement, and total CEV vs the no-ETS baseline SS.

5.1 A welfare-optimal permanent carbon tax

Before comparing ex ante and ex post, we characterise the steady state a permanent τ_b delivers. Holding ψ_g fixed at its no-tax value and letting D_{ss}^G float, a carbon tax raises the effective brown/green price ratio and lifts steady-state adoption: D_{ss}^G moves from 5% at $\tau_b = 0$ to 8.9% at $\tau_b = 0.10$ and 20.1% at $\tau_b = 0.30$ (Figure 8, right). The carbon revenue is rebated lump-sum, so the exercise is budget-neutral and isolates the allocative effect.

Steady-state welfare is hump-shaped in τ_b (Figure 8, left): the standing CEV rises to a peak of +0.043% at $\tau_b^* \approx 10\%$ and turns negative beyond $\tau_b \approx 25\%$. The model contains *no* climate externality, so this interior optimum is not Pigouvian. It is a second-best correction of the adoption friction: green energy is structurally cheaper ($\bar{P}_G^E = 0.8 < \bar{P}_B^E = 1$) but the logit taste-shock scale σ_ε and the switching cost ψ_g leave green under-adopted, so a modest tax that pushes households toward the cheaper technology raises welfare until the tax distortion dominates. That the optimum is positive at all is a property of this friction, and we treat it as a diagnostic of the channel rather than a policy recommendation—it would be swamped by any explicitly modelled externality or abatement cost.

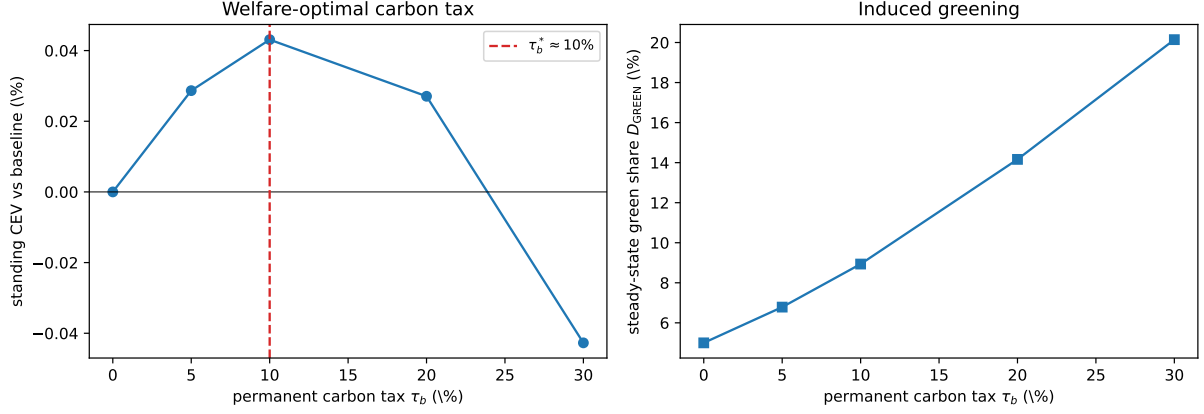


Figure 8: The welfare-optimal permanent carbon tax. Left: standing CEV of the τ_b steady state relative to the no-tax baseline (no crisis), hump-shaped with an interior optimum near $\tau_b^* \approx 10\%$. Right: the induced steady-state green share D_{ss}^G . ψ_g is held fixed and D_{ss}^G floats; the carbon revenue is rebated lump-sum.

5.2 Ex ante greening barely insures against the crisis

We now hit two economies with the *same* price shock: the baseline ($D_{ss}^G = 5\%$) and a “prepared” economy carrying the welfare-optimal $\tau_b = 0.10$ ($D_{ss}^G = 8.9\%$). Table 6 reports the total CEV of each, alongside the ex post cap and transfer on the baseline economy.

Scenario	CEV%	impat.	middle	patient	peak D_G	gross fisc.
Laissez-faire	-1.871	-2.370	-2.048	-1.196	8.87	0.00
Ex-post cap	-1.147	-1.049	-1.253	-1.138	-0.04	53.86
Ex-post transfer	-0.816	-0.682	-0.886	-0.880	9.15	49.27
Ex-ante ETS	-1.852	-2.294	-2.016	-1.246	14.98	8.22
Ex-ante ETS + cap	-1.087	-0.942	-1.180	-1.137	-0.06	61.65

Table 6: Ex-ante greening vs ex-post capping under a brown-energy price shock (import booking, $\tau_b = 0.1$). CEV is the total consumption-equivalent variation relative to the no-ETS steady state, by discount-factor type; gross fiscal is the cap outlay (ex-post) or the standing carbon revenue over $H = 24$ (ex-ante).

The prepared economy is essentially no better insured than the unprepared one: its crisis CEV is -1.85% against -1.87% for laissez-faire, a difference of two basis points, and both are far worse than either ex post instrument (cap -1.15% , transfer -0.82%). Adding the standing benefit of the tax ($+0.04\%$, Section 5.1) does not change the ranking. The greening the tax buys is real— D_{ss}^G is 78% higher—but too small a share of the economy to move aggregate exposure appreciably, and the tax’s crisis-time value is a rounding error next to a transfer that puts cash in high-MPC hands exactly when it binds. Consistent with Section 4.3, the combination “ETS + cap” tracks the cap alone: once the price is frozen the adoption margin is shut and the greener starting point is irrelevant.

The green *subsidy* of Section 4.3¹ is worth a line as a crisis instrument in its own right. It doubles the adoption response (peak D^G of 18.0 vs. 8.9 pp) at trivial fiscal cost, yet its CEV is the worst in the table, -2.03% , below laissez-faire. Paying households to incur the real switching cost $\psi_g D^{sw}$ —booked as an import (Section 2.6)—in the middle of a consumption crisis destroys

¹The transitory switching subsidy s_g , financed by debt over the crisis window.

welfare even as it greens the economy. The adoption margin is valuable, but not as something to force at the trough.

5.3 Persistence exhausts the adoption absorber

The obvious defence of the ex ante case is persistence: a longer-lived shock should let the greener economy’s lower exposure compound over more periods. Figure 9 and Table 7 test it by sweeping the shock half-life and re-solving only the impulse response (the steady state does not move with persistence), decomposing the ex ante number into its constant standing part and a *crisis-only greening dividend*, defined as the prepared economy’s crisis CEV minus the standing +0.04% minus laissez-faire.

half-life	CEV _{LF}	CEV _{cap}	CEV _{ETS}	green div.	cap prot.
4	-0.648	-0.406	-0.594	0.0110	0.242
8	-1.158	-0.725	-1.112	0.0022	0.432
12	-1.553	-0.963	-1.520	-0.0108	0.590
16	-1.871	-1.147	-1.852	-0.0238	0.724
24	-2.358	-1.414	-2.361	-0.0462	0.944
32	-2.717	-1.602	-2.737	-0.0639	1.115
48	-3.214	-1.850	-3.260	-0.0891	1.364

Table 7: Route A: welfare by shock persistence (import booking, $\tau_b = 0.1$). Total CEV vs the no-ETS baseline SS. The crisis-only greening dividend (CEV_{ETS} minus the standing +0.043% minus CEV_{LF}) does not grow with persistence, while the cap protection does.

The defence fails, and in an instructive way. The greening dividend does not grow with persistence—it *turns negative*, from +0.01% at a four-quarter half-life to −0.09% at a twelve-year one, crossing zero near a three-year half-life. The ex post cap moves the other way: its crisis protection rises monotonically, from +0.24% to +1.36%, as a longer freeze shields more of the transition.

The reason is the adoption margin itself. A persistent brown-price shock makes switching attractive for many quarters, so the *browner* laissez-faire economy adopts heavily *during* the crisis and reaps that windfall over the whole episode—the margin is an endogenous shock absorber. The prepared economy has already spent much of that absorber in the steady state: with fewer brown households left at the switching margin, it has less adoption to do when the shock arrives, and less of the crisis-time windfall to collect. Pre-greening does not merely fail to insure against a persistent shock; it *consumes the very margin* that would have cushioned it. This is the sharpest form of the paper’s thesis. The instruments that freeze the adoption technology—the cap here, and by construction the frozen-technology models of Bayer et al. (2026), Langot et al. (2026) and Auclert et al. (2024)—miss an absorber that works best when left loaded, i.e. when the economy enters the crisis with room to green rather than already green.

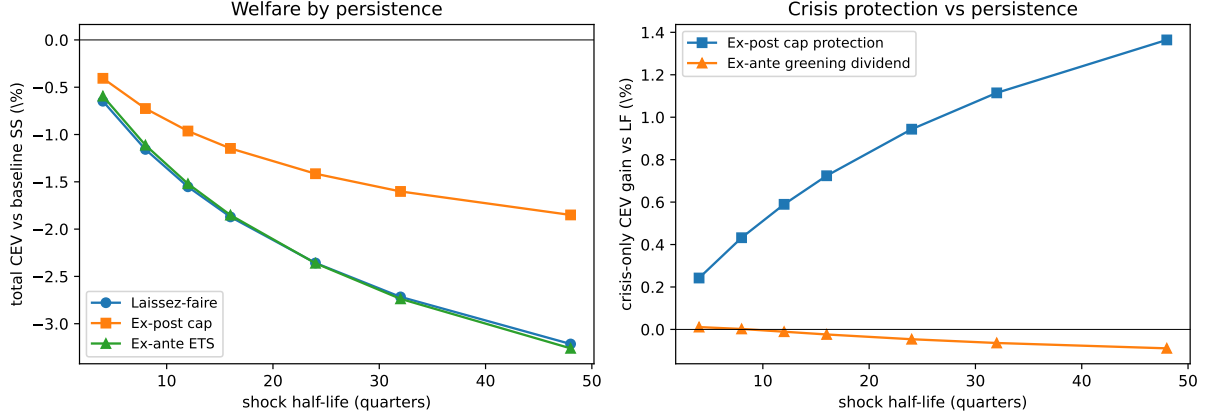


Figure 9: Persistence and the two policies. Left: total CEV of laissez-faire, the ex post cap and the ex ante ETS as the shock half-life rises. Right: the crisis-only gain over laissez-faire—the cap’s protection grows with persistence, while the ex ante *greening dividend* turns negative, because a more persistent shock lets the browner economy exploit the adoption absorber the prepared economy has already spent.

Takeaway. Three results point the same way. Energy-indexed targeting is dominated by a flat transfer of equal budget (Section 4.5); a permanent carbon tax is mildly welfare-improving in the steady state through a non-Pigouvian adoption correction, but essentially worthless as crisis insurance (Sections 5.1–5.2); and its insurance value turns negative as shocks become persistent, because greening ex ante exhausts the adoption absorber (Section 5.3). The margin this paper adds is not a lever a planner should pre-pull; it is a stabiliser that pays off precisely when the economy is left free to adopt in the crisis.

6 Relation to the literature

This section is written for the coauthors: it states, paper by paper, what our setup changes relative to the reference and what that change does to the results. Compact one-page summaries of each reference are in Appendix C.

6.1 Relative to Auclert et al. (2024)

Setup. We inherit the entire ARS aggregate economy—the CES energy nest, sticky import prices, the mutual fund, the energy suppliers with intertemporal arbitrage, the real-wage-stabilising wage Phillips curve, and the constant-real-rate monetary baseline—and change the household in four coupled ways. (i) ARS’s household is homothetic with a fixed energy technology: every household shares one CES basket and one CPI, and energy demand is $\alpha_E (P_E/P)^{-\eta_E} c$. We add a discrete brown/green durable state and a DC-EGM adoption choice, so the energy technology becomes an endogenous, household-level object. (ii) ARS has a single household energy price; we have two (P_B^E, P_G^E) and a type-specific cost-of-living index. (iii) ARS uses the CPI as unit of account; we use the domestic good, so that the resource constraint is invariant to how the CPI weights brown and green energy. (iv) We add two current-account terms absent in ARS—the switching cost as an import, the adopters’ energy saving as an import saving. The energy-supply closure is a generalisation of ARS: their baseline is our $\varepsilon^E = \infty$ price-taking case, and their coordinated world equilibrium (endogenous world price) is the two-country analogue of our ε^E -finite case.

Results. ARS’s central mechanism—low χ makes the energy price shock contractionary through the real-income channel—is preserved and operates in the background of every one of our experiments. Where we depart is that ARS’s fiscal ranking is entirely about output, inflation and inequality: in ARS the subsidy is a domestic “silver bullet” that stabilises all three, and its only drawback is a *cross-country* externality (coordinated subsidies spike the world price and become self-defeating). Our model adds an orthogonal cost of the same subsidy that ARS cannot see: capping P_B^E eliminates the green adoption margin (peak D^G from 9.18 to 0.01 pp), so the subsidy is not neutral for the transition. Two further connections are worth flagging to the coauthors. First, our supply-shock result—the cap is destructive under inelastic supply—is the *single-country* counterpart of ARS’s coordination result: in both, capping the price under an endogenous/scarcie energy price prevents the necessary substitution and deepens the recession. Second, our monetary baseline being ARS’s constant-real-rate rule is what pins the real exchange rate, so a domestic transfer has no terms-of-trade effect; this is inherited, not assumed anew.

6.2 Relative to Bayer et al. (2026)

Setup. Bayer et al. is a two-country HANK monetary union (Germany one-third, rest of the union two-thirds), with a fixed nominal exchange rate, built to study *spillovers*. Our economy is a single small open economy with an exogenous foreign block, so by construction we cannot represent their central object—the cross-border, beggar-thy-neighbour transmission of a national subsidy. Three further setup differences matter. (i) Supply: Bayer fix the union-wide energy *quantity* and let a common price clear; this is exactly our ε^E -finite closure, so our supply-shock experiments are the natural comparison. Our shock is smaller (a 10% cut for 6 quarters against their 20% cut for 6 quarters), chosen for first-order reliability. (ii) Supply side: Bayer has a rich medium-scale block—two assets (liquid bonds, illiquid capital), an investment margin with capacity utilisation, energy in production as well as consumption, and GHH preferences. We have a single asset, no capital, and energy in consumption only in the baseline. (iii) Heterogeneity: Bayer’s headline distributional mechanism is *within-country energy-share heterogeneity*—poor households have systematically higher energy shares (a high/low-intensity Markov type correlated with income), so they suffer larger real-income losses. We assume homothetic energy demand conditional on the durable type, so we cannot speak to their energy-exposure distributional results; our heterogeneity is over discount factors and over the adoption margin instead. Finally, their monetary policy is a standard Taylor rule, not the constant-real-rate rule; they show this matters, because under a Taylor rule the subsidy also suppresses *intertemporal* substitution through household-specific real rates.

Results. Under the supply closure we reproduce the qualitative core of Bayer et al.: the Slutsky/Hicks transfer welfare-dominates the price cap (our private CEV -0.16 vs. -1.28), and capping the price under a fixed quantity is inefficient because it prevents the substitution the scarcity requires—our “cap pushes towards Leontief” is their “subsidising under a supply contraction is inefficient.” What we add is the adoption margin, which Bayer et al. do not have: the same cap that is inefficient for output also switches off the transition, and the transfer that is welfare-superior also preserves it. What we *cannot* deliver is their spillover result: with no second country, we have nothing to say about whether one country’s subsidy harms its neighbours. This is the reason we are careful (Section 4.3, and the abstract) not to claim that a single parameter “reconciles” Bayer and Langot; we reproduce the domestic sign, not the cross-border mechanism.

6.3 Relative to Langot et al. (2026)

Setup. Langot et al. study France as a single economy inside the euro area, with an ECB Taylor rule down-weighted for France’s size. Two structural choices distinguish their model from ours. (i) They take the energy price as *exogenous* (fed from the government’s official

forecasts), so France is a price taker facing a perfectly elastic supply at that price—this is our $\varepsilon^E = \infty$ closure, and their price-shock scenario is the right comparison. (ii) Their energy substitution elasticity is $\eta_E = 0.5$, five times ours, and they add an *incompressible* (subsistence) level of energy consumption, which delivers non-homotheticity: poor households have a larger incompressible share, hence a higher effective energy share and a lower price elasticity. This is their distributional mechanism, analogous in spirit to Bayer’s energy-share heterogeneity but generated by subsistence rather than by types. Our model buys energy substitution on a different margin: we keep the *intensive* elasticity low ($\eta_E = 0.1$, as in ARS) and obtain substitution from the *extensive* adoption margin instead. A third, methodological difference matters for how the results should be read: Langot et al. conduct an *ex-ante, conditional-forecast* evaluation—they back out the sequence of structural shocks that makes the HANK reproduce the French government’s published forecasts for output, inflation and debt, then run counterfactual policies holding those shocks fixed. Ours (like ARS and Bayer) is a clean MIT-shock impulse-response exercise, not a fit to a specific historical episode.

Results. Under the elastic (price) closure we agree with Langot’s headline: the cap cushions output ($\sum y = -20.9$ vs. -30.1 with no policy) and, on private welfare, tightening the cap raises the CEV ($-1.30 \rightarrow -0.57$)—consistent with their finding that the shield supports growth and curbs inflation. But two qualifications are important for the coauthors. First, Langot’s pro-shield result rests substantially on a *labour-cost* channel: by containing the retail energy price the shield prevents the price–wage spiral, holds down the cost of labour, and thereby supports employment; in their comparison the shield beats both wage indexation and a targeted transfer precisely on this employment margin. Our wage block and our lack of an investment/employment margin mean we reproduce the *sign* of their result but not the full mechanism—hence we do not claim the Langot–Bayer contrast is “only” about the supply elasticity. Second, Langot et al. hold the energy technology fixed, so like ARS and Bayer they do not see the cap’s cost to the transition, which is our contribution.

7 Limitations

1. **The taste scale σ_ε is calibrated, not estimated.** ψ_g and σ_ε jointly hit $D_{ss}^G = 0.05$, but one level moment cannot separately identify both a switching cost and a taste scale: a second moment (e.g. the semi-elasticity of adoption to a purchase subsidy) is needed and would discipline the sign found in Section ?? along with the magnitudes throughout. The *ordering* of the cap-vs-transfer results does not depend on it, since it follows from the relative price alone.
2. **The switching cost is booked as an import by accounting necessity.** Under ARS’s $\mu_{ss} > 1$ with capitalised profits, booking $\psi_g D^{sw}$ as domestic demand creates a marginal dividend owned by nobody and breaks asset-market clearing; we therefore book it externally (Section 2.6). This convention is what makes the adoption channel *contractionary* under the price shock, since the cost lands in the periods household income is falling hardest (Section ??). The *ordering* of the policy results does not depend on it—it follows from the relative price alone—but the sign of the adoption channel’s output contribution under a price shock does, and should be read as conditional on this booking.
3. **Zero pass-through to the green price ($\rho_g = 0$)** means adopters are completely insulated from the fossil shock: an upper bound on the adoption channel. A partial pass-through calibrated to observed electricity–gas co-movement would be more defensible.
4. **No modelled green technology.** The adopters’ resource saving rests on the domestic/imported interpretation of Section 2.6 rather than on a green sector with its own ca-

capacity, cost curve and investment. We have implemented the minimal version—a competitive, zero-profit installer sector that books the switching cost as domestic value added rebated to households—and it clears to machine precision and flips the sign of the adoption channel (Section 4.2); what remains unmodelled is the green energy *supply* technology (capacity, cost curve, investment) and a distribution of installer income more realistic than the lump-sum rebate.

5. **No within-country energy-share heterogeneity.** Bayer et al. (2026)’s central distributional mechanism is that poor households have higher energy shares; we have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.
6. **No investment margin.** Both Bayer et al. (2026) and Auclert et al. (2024) have richer supply sides. Its absence here likely makes the transfer look more expansionary, and the supply shock more inflationary, than it would be with capital to smooth the adjustment.

A Balance-of-payments booking of the adoption flows

The sign and magnitude of the adoption channel’s contribution to output depend on how the green-margin resource flows—the switching cost $\psi_g D^{sw}$ and the green energy bill—are booked in the balance of payments. The main text reports the import booking throughout. This appendix collects the two results that hinge on that choice: the domestic-booking sign comparison, and the carbon-revenue recycling experiment, which is naturally a booking-conditional statement about where the switching subsidy lands.

A.1 Import versus domestic booking of the adoption flows

Both statics of Section 4.2 are conditional on the import booking of the adoption flows. We solve the model under the alternative domestic booking of Section 2.6—green energy and green installation as competitive, zero-profit domestic value added rebated to households, with only brown energy imported—which resolves the phantom-dividend problem and clears to machine precision. Under this booking neither the switching cost nor the green energy bill is a leakage, and the adoption channel’s contribution to cumulative output (Δ_{24}) is as following:

Δ_{24} , adoption output contribution	import booking	domestic booking
price shock	−1.60	+35.98
supply shock	+0.38	+5.96

Table 8: Adoption channel contribution to cumulative output (Δ_{24}), import vs. domestic booking (Section 2.6). *None* policy.

A.2 Recycling the carbon revenue: rebate versus green subsidy

The welfare-optimal carbon tax of Section 5.1 returns its revenue lump sum. An alternative earmarks it to the transition itself, paying a share s_g of the green switching cost. We compare the two recyclings across carbon-tax rates $\tau_b \in \{0.10, 0.20, 0.30\}$, holding ψ_g at its no-tax value so the steady-state green share floats (Figures 10 and 11). Under the lump-sum rebate the steady-state green share rises from 8.9% to 20.1% across the grid, and the whole brown-price path in the crisis shifts up by the factor $(1 + \tau_b)$: the pre-tax price passes the world shock through identically, while households pay the tax-inclusive price. Under *budget-neutral* green-subsidy recycling— s_g solved so that the switching subsidy exactly exhausts the carbon revenue ($T^{\text{reb}} = 0$)—the revenue funds a subsidy covering 35–40% of the switching cost and lifts the steady-state green share to 27–46%.

Two features are worth flagging. First, the carbon base is self-eroding: at $\tau_b = 0.10$ the revenue is a third smaller under green-subsidy recycling than under the rebate, because the greener economy holds less brown-durable stock to tax. Second, the crisis-time greening response is *smaller*, not larger, from the greener starting point under the subsidy: recycling that greens the economy through a lower switching *cost* preferentially converts the households nearest the logit margin, leaving a residual brown pool that is less responsive to the shock—the mirror image of the rebate, which greens through a higher permanent brown *price* and leaves the residual pool closer to the margin. This is the adoption-absorber exhaustion of Section 5.3 seen across policy instruments rather than across shock persistence.

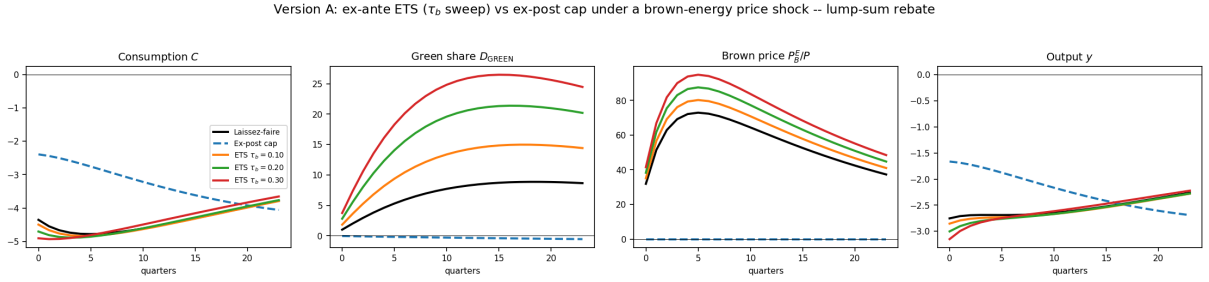


Figure 10: Carbon-tax sweep, lump-sum rebate: laissez-faire and ex-post cap against the ex-ante ETS economy at $\tau_b \in \{0.10, 0.20, 0.30\}$.

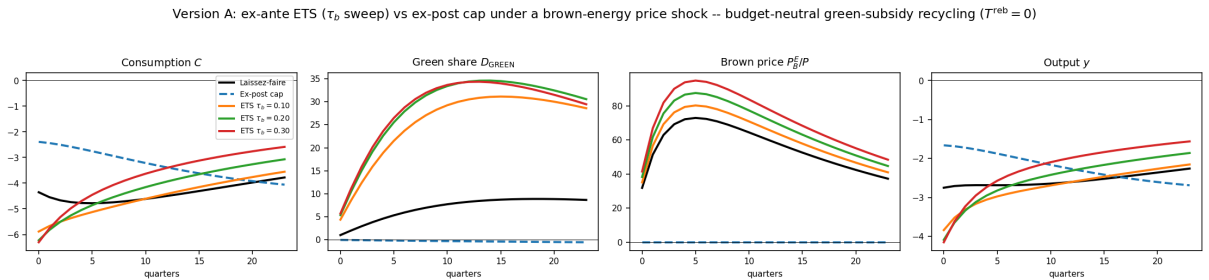


Figure 11: Carbon-tax sweep, budget-neutral green-subsidy recycling ($T^{\text{reb}} = 0$, s_g endogenous). Same axes as Figure 10.

B The linearised solution and the CPI deflator

Two robustness exhibits that support the main-text results without being part of the argument: the accuracy of the linearised (sequence-space) solution given the discrete adoption choice, and the measurement gap in the brown-anchored CPI.

B.1 Robustness: the nonlinearity lives in the adoption margin

A natural concern is that the discrete adoption choice makes the economy too nonlinear for the linearised (sequence-space) solution to be reliable, in contrast to Auclert et al. (2024), who find nonlinearity immaterial in the continuous economy. Comparing the linearised impulse responses with their nonlinear counterparts (a fixed-Jacobian Newton solve) shows the concern is confined to one variable. Aggregate output is essentially linear: the ratio of the nonlinear to the linear peak response, $y^{\text{NL}}/y^{\text{L}}$, stays in $[1.00, 1.01]$ across shock sizes, reproducing the small nonlinearity ARS report. The nonlinearity is concentrated in the adoption share D^G , whose ratio rises from

1.16 at a one-eighth shock to 2.04 at a three-quarter shock (Figure 12, left). With the margin shut, the bare household block is in fact mildly *concave* ($y^{\text{NL}}/y^{\text{L}} \approx 0.92$ at the largest shock); the adoption margin is *convex* and partly offsets it, which is why output is, if anything, more linear with the margin open. **The "margin shut" comparison in this subsection predates the common-steady-state counterfactual** (Section 4.1) and uses the OLD, zero-steady-state construction, which is arguably still the right comparison for THIS specific exercise (isolating the bare continuous-HANK block's own curvature, to compare against ARS's technology-free economy, is a different question from isolating the crisis response of the discrete choice); but for consistency with the rest of the paper, and because `model.py` no longer exposes that construction under `model_variant='no_adoption'`, this comparison should be redone against the frozen (common-SS) counterfactual before submission, with an explicit note on which of the two questions is being asked.

The convexity is the intrinsic curvature of a logit share evaluated near a low level. For a binary choice $P = [1 + \exp(-\Delta/\sigma_\varepsilon)]^{-1}$ in the value gap Δ , the relative curvature is $(1 - 2P)/\sigma_\varepsilon$, so $D^{G,\text{NL}}/D^{G,\text{L}} - 1 \propto 1/\sigma_\varepsilon$. Re-solving on a grid of σ_ε (holding $\bar{D}^G = 0.05$ by recalibrating ψ_g) confirms this: the ratio falls monotonically from 2.21 at $\sigma_\varepsilon = 0.02$ to 1.06 at $\sigma_\varepsilon = 0.40$ (Figure 12, right), and it tends to one as the shock vanishes, so the linearisation is exact to first order and the Jacobian is correct. Two implications follow. Aggregate and consumption-based welfare statistics are reliable under linearisation. The *level* of induced greening at the largest shocks is understated by the linear solution—and does not converge under a full-size shock—so adoption-channel magnitudes are read at a reduced shock size, and the taste scale σ_ε that governs this curvature is the same parameter left underidentified in Section 7: an adoption-elasticity moment would pin both at once.

E9. The nonlinearity lives in the discrete adoption margin

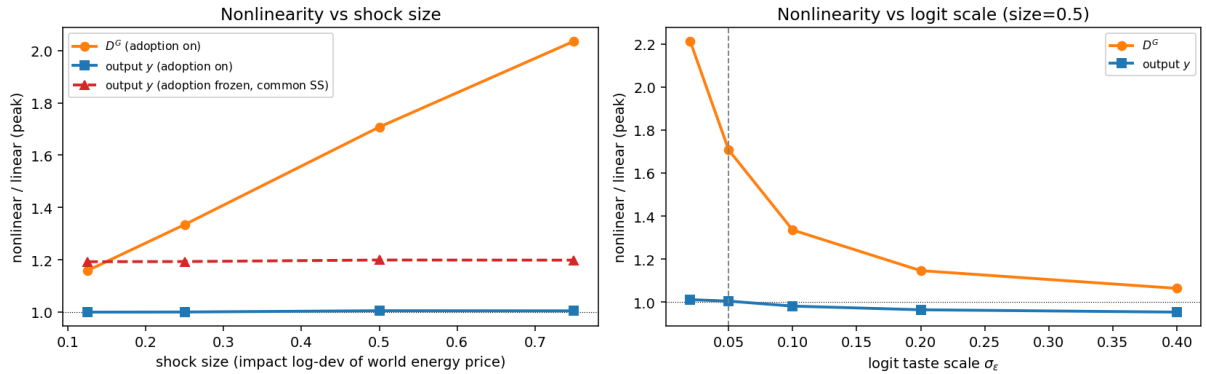


Figure 12: The nonlinearity is confined to the discrete adoption margin. *Left*: peak nonlinear/linear ratio vs shock size—output (adoption on and off) and D^G . *Right*: the same ratio vs the logit taste scale σ_ε at a half-size shock, ψ_g recalibrated to $\bar{D}^G = 5\%$.

B.2 The brown-anchored CPI understates the cost of living

The model's CPI anchors its energy leg on the brown price alone (eq. 32), while the economically correct index reweights brown and green by the adoption share. The gap has the closed form $\phi^{1-\eta_E} = 1 + \alpha_E D^G [(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}]$, computed ex post so it closes no loop in the model. It is small but not zero: the measured deflator overstates the cost of living by at most a few tenths of an annualised percentage point over the crisis, and the overstatement grows with adoption (Figure 13). Because the monetary rule reads the measured CPI, this is the wedge a Taylor rule would miss; under the domestic-good numeraire it affects only the measured deflator and the rule, not the resource constraint.

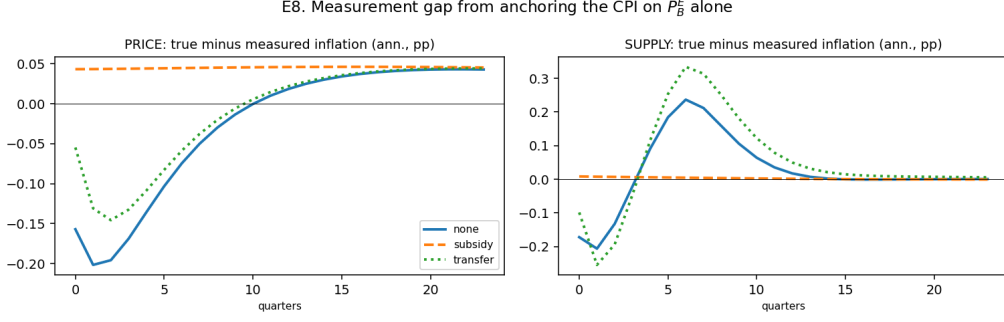


Figure 13: True minus measured annualised inflation from anchoring the CPI on P_B^E alone, by policy.

C Summaries of the three reference papers

These summaries are in our own words, for internal reference. They are deliberately compact—roughly one page each—and emphasise the features that matter for our comparison in Section 6.

C.1 Auclert et al. (2024), “Managing an Energy Shock”

Auclert et al. (2024) ask how an energy-importing economy should respond to an energy price shock, and whether household heterogeneity changes the answer. The model is a small open economy HANK built on the exchange-rate model of Auclert, Rognlie, Souchier and Straub (2021), extended with an energy good. There are three goods: a domestically produced home good, an imported foreign good, and imported energy. Households face idiosyncratic productivity risk, cannot insure it, and hold a single asset through a mutual fund; preferences are CRRA in a CES consumption basket that nests energy against a home/foreign bundle, with a low energy substitution elasticity ($\eta_E = 0.1$) and a total substitution index $\chi = 0.3$. Crucially, all households share the *same* basket and the *same* price index—the energy technology is fixed and homothetic. Wages are sticky with a real-wage-stabilisation motive; the monetary baseline is a rule that holds the real interest rate constant. Energy is supplied by price-taking foreign firms with an intertemporal extraction margin, so in the single-country equilibrium the world energy price is exogenous.

The analytical core is a decomposition of the output response into an expenditure-switching channel (positive) and a real-income channel (negative) scaled by the matrix of intertemporal MPCs. In the complete-markets representative-agent benchmark only expenditure switching survives, so the shock is expansionary; with realistic MPCs and low χ the real-income channel dominates and the shock is contractionary—their central result, and the one we inherit. A neutrality result holds at $\chi = 1$, where the two channels cancel. With real wage rigidity a wage–price spiral can appear, but the real wage still falls.

On policy, ARS study monetary tightening and three fiscal tools—energy subsidies, targeted transfers (indexed to counterfactual energy use), and untargeted transfers, all deficit-financed and repaid by a labour tax. For a country acting alone, subsidies are the most effective domestic stabiliser and tame inflation (they cap the price that drives the real-wage loss); transfers support demand but are more inflationary. The decisive twist is *cross-country*: monetary tightening has a positive externality (it lowers world energy demand and price), while subsidies have a large negative externality (coordinated subsidies make world energy demand nearly inelastic, spike the world price, and become self-defeating). Their policy recommendation is coordinated tightening plus transfers targeted to the poor, avoiding subsidies. They also show the impulse responses are not strongly nonlinear at empirically relevant shock sizes.

For our purposes the key features are: fixed homothetic energy technology, a single energy price and CPI, the constant-real-rate monetary rule, and the fact that the only “cost” of a subsidy in their world is a cross-country externality—there is no transition margin.

C.2 Bayer et al. (2026), “Hicks in HANK” / “Redistribution Within and Across Borders”

Bayer et al. (2026) evaluate the fiscal response to the European energy crisis in a two-country HANK model of a monetary union, calibrated with Germany as “Home” (one-third of the union) and the rest of the area as “Foreign” (calibrated to Italy). The model is a medium-scale, two-asset HANK: households hold a liquid union-wide bond and illiquid capital, face idiosyncratic income risk, and have GHH preferences; there is an investment margin with capacity utilisation. Energy enters both production (a CES bundle with the capital–labour aggregate, elasticity 0.2) and household consumption (a CES bundle with the physical good, elasticity 0.1). A distinctive and, for them, central feature is within-country heterogeneity in the *energy share*: households are of high- or low-energy-intensity type, following a persistent Markov process correlated with income, so that poorer households spend a larger share on energy and suffer larger real-income losses. Monetary policy is a standard union-wide Taylor rule; each national fiscal authority taxes labour and profits and stabilises its debt.

The defining assumption is that the total union energy supply is *inelastic*: import capacity is fixed in the short run, so a fixed quantity is set exogenously and a common price clears the market. The crisis is a 20% cut in supply for six quarters, in response to which the energy price rises roughly fivefold, inflation rises about four points, and output falls around one percent, with consumption falling most for poor, high-energy households.

They compare two policies, each implemented in Home only so that spillovers can be measured: a subsidy that caps the retail energy price at its pre-crisis level, and a Hicks/Slutsky transfer indexed to pre-crisis energy consumption (to households and firms). The subsidy is the more effective domestic stabiliser but is fiscally expensive and *beggar-thy-neighbour*: by holding Home’s energy demand up it pushes the common price higher, amplifying the crisis in Foreign. The transfer stabilises less at Home but generates essentially no spillover and is cheaper. On welfare (consumption-equivalent variation), the subsidy *reduces* Home welfare—subsidising energy under a supply contraction is inefficient because it suppresses both intratemporal and intertemporal substitution—while the transfer *raises* Home welfare, its cost being mostly the distortion of the taxes that finance it. Their headline contrast with ARS (subsidy more stabilising than transfer for output) they attribute to the Taylor rule and the two-country structure, versus ARS’s constant-real-rate rule.

For our purposes the key features are: fixed union-wide quantity (our inelastic closure), the two-country spillover mechanism that we cannot represent, the within-country energy-share heterogeneity that we do not have, and the substitution-preserving property of the transfer, which we share.

C.3 Langot et al. (2026), “The Macroeconomic and Redistributive Effects of the French Tariff Shield”

Langot et al. (2026) evaluate the French *bouclier tarifaire*—the 2022–23 cap on gas and electricity prices—in a HANK model of France as a single economy within the euro area. The model extends Auclert, Rognlie, Souchier and Straub with energy in both household consumption and production. Households face idiosyncratic productivity risk and hold a single asset; preferences are CRRA with a CES energy nest, and the energy substitution elasticity is $\eta_E = 0.5$, markedly higher than in ARS or Bayer. A distinctive feature is an *incompressible* (subsistence) level of energy consumption: only energy above the subsistence level yields utility. This generates non-homotheticity without non-homothetic preferences—poorer households devote a larger share to

incompressible energy and have a lower price elasticity of energy demand, which is the source of the distributional results. The ECB Taylor rule is down-weighted to reflect France’s small weight in euro-area inflation. In the policy simulations the energy price is taken as exogenous (from the government’s forecasts), so France is effectively a price taker facing perfectly elastic supply.

The methodology is distinctive and worth flagging: rather than clean MIT-shock impulse responses, the paper performs an *ex-ante, conditional-forecast* evaluation. Using the sequence-space Jacobian, the authors back out the sequence of structural shocks (energy price, government spending, transfers, plus latent demand, markup and fiscal shocks) that makes the model reproduce the French government’s official published forecasts for output, inflation and the debt ratio through 2027. Policies are then evaluated as counterfactuals that hold this shock sequence fixed and change one instrument.

The main finding is that the tariff shield supports growth (about 1.9% average over 2022–23 against 1.0% without it) and curbs inflation (5.6% against 7.2%), at a fiscal cost near 2% of GDP per year and a modest rise in the debt ratio, while containing—though not eliminating—the rise in consumption inequality even though it is untargeted. The mechanism behind its effectiveness is a *labour-cost* channel: by holding down the retail energy price the shield prevents the price–wage spiral, contains the cost of labour, and thereby supports employment and output. On this macro criterion the shield beats both a faster wage indexation (more inflationary, worse for output) and a targeted transfer to finance incompressible energy consumption; the targeted transfer, however, reduces inequality by more.

For our purposes the key features are: exogenous/elastic energy price (our price-taking closure), the labour-cost channel behind the pro-shield result (which our wage block only partly captures), the higher intensive substitution elasticity, and again a fixed energy technology with no transition margin.

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